

Premises: 1) $T \rightarrow (M \vee E)$ 2) $P \rightarrow \neg E$ 3) $T \wedge P$

Conclusion: M

Clauses: 1) $C_1: \neg T \vee (M \vee E)$ 2) $C_2: \neg P \vee \neg E$ 3) $C_3: T$
4) $C_4: P$ 5) $\neg M \rightarrow C_5$

By resolution from C_1 and C_3 , we get $C_6: (M \vee E)$

By resolution from C_2 and C_4 , we get $C_7: \neg E$

By resolution from C_6 and C_7 , we get $C_8: M$

By resolution from C_8 and C_5 , we get $C_9: \square$

Hence, the argument is correct. $\square$

1.6.6. Logic Programming

An important type of programming language is designed to reason using the rules of predicate logic. Prolog (from Programming in Logic), developed in the 1970s by computer scientists working in the area of artificial intelligence, is an example of such a language.

Prolog programs include a set of declarations consisting of two types of statements, Prolog facts and Prolog rules. Prolog facts define predicates by specifying the elements that satisfy these predicates. Prolog rules are used to define new predicates using those already defined by Prolog facts.

Example 12: Consider a Prolog program given facts telling it the instructor of each class and in which classes students are enrolled. The program uses these facts to answer queries concerning the professors who teach particular students. Such a program could use the predicates $\text{instructor}(p, c)$ and $\text{enrolled}(s, c)$ to represent that professor p is the instructor of course c and that student s is enrolled in course c , respectively.